

JPE Replication Report

PUBLISHED

July 10, 2026

1 Paper Identification

ID	2016256
Author	Mitman
Title	Unemployment Benefits and Unemployment in the Great Recession: The Role of Macro Effects
Iteration	1

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. Please add correct data citations in both readme and paper. We have specific [guidance here](#). In the paper, please add one citation per data set (not one citation per vintage).
2. We found a few discrepancies and we are unsure whether those arise from not having access to the HWOL data or not. From your code it seems you intended to give us temporary access to those data (this would be much preferred); we did not find any confidential data in the package. Please provide a table in the readme which explicitly lists all exhibits which can and which can not be reproduced without access to the confidential data.
3. The code mistakes we found are minor, but maybe easy to fix.
4. Please provide info about any random seeds that need to be set.
5. Regarding access to the confidential data, please describe in detail all the necessary steps in order to obtain the data - this is true even though you might consider such data as fairly standard in your field.
6. I need a dedicated Data Availability and Provenance Statement in your readme.
7. Please include both statements about rights below.

3 General

☐ Data Availability and Provenance Statements

- ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
- ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
- ☐ License for Data (optional, but recommended)
- ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☐ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
BLS LAUS	yes	no	yes	yes	no	no
BLS JOLTS	yes	no	yes	yes	no	no
BLS QCEW	yes	no	yes	yes	no	no
Census LEHD QWI	yes	no	yes	yes	no	no
Census LODES	yes	no	yes	yes	no	no
BEA state GDP	yes	no	yes	yes	no	no
BEA CA25N industry employment	yes	no	yes	yes	no	no
FHFA HPI	yes	no	yes	yes	no	no
IRS SOI county income (via replication archive)	yes	no	yes	yes	no	no
NY Fed / Equifax county debt-to-	yes	no	yes	yes	no	no

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
income						
Saiz (2010) housing supply elasticity and WRLURI	yes	no	yes	yes	no	no
Census state tax revenue	yes	no	yes	yes	no	no
State business- policy indices	yes	no	yes	yes	no	no
USDA SNAP policy database	yes	no	yes	yes	no	no
Recovery.gov ARRA awards + geocorr crosswalk	yes	no	yes	yes	no	no
Census population estimates	yes	no	yes	yes	no	no
Shimer CPS separation series	yes	no	yes	yes	no	no
DOL / state UI benefit policy triggers	yes	no	yes	yes	no	no
County UI claims + UI tax receipts	yes	no	yes	yes	no	no
Census geography	yes	no	yes	yes	no	no
Conference Board HWOL vacancies (proprietary; synthetic stand- in provided)	no	yes	yes	yes	no	no

Note: These are not formal data citations; the data citations should look like paper citations and be present in references in a consistent format. Please correct them, include year of the dataset and the access date. Add (permanent) urls.

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - **Stata/MP** (the build and the `egen`-heavy steps assume MP). All required user-written commands install **automatically from SSC**: `code/config.do` — which every pipeline script sources — checks for and installs `regife` (interactive fixed effects, the Stata-side IFE cross-checks), `reghdfe`, `ivreg2` (+ its dependency `ranktest`), `outreg2`, `binscatter`, `ftools`, `coefplot`, `tuples`, `hdfe`, and `mat2txt` on first run (an internet connection is needed once). No packages are bundled in the archive; in particular `regife` is **not** redistributed — it loads from SSC like the rest.
 - **MATLAB** (developed on R2025b; the factor model and bootstrap reproduce across MATLAB versions) for the factor model, block bootstrap, and the structural model.
 - **Python 3** for `code/tex/make_tables.py` (standard library only) and a TeX distribution to compile the paper.
- ☒ Computational Requirements specified as follows:
 - Peak disk while running: the built `data/processed/` master datasets are ~5 GB; the shipped archive itself (raw + provided inputs + code, no built files) is smaller.
- ☒ Time Requirements specified as follows:
 - The Stata build (`data/processed` from `data/raw`) runs in a few minutes on a multi-core machine.
 - The factor-model **block bootstrap** (~200 reps over ~1,100 county pairs) is the expensive step — **hours** per specification, and there are ~30 of them; the full `do_rep.sh` run is on the order of 10–15 hours on a multi-core machine. The bootstrap result CSVs are written to `output/factor_results/` as the run proceeds (the archive ships no pre-computed outputs).
- ☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 8.2](#)

7 Analyzing Data and Code Files

7.1 Code description

- ✓ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

7.1.1 Summary of output-producing code

The package contains a master code in [replication-package/JPEReplicationKitFinal/code/RunBuild.do](#) that runs the raw-data build steps and the processed-data construction stages. The paper tables are assembled by [replication-package/JPEReplicationKitFinal/code/tex/make_tables.py](#), which reads CSV outputs generated by the Stata and Matlab analysis scripts and writes .tex fragments into the output/tables directory. The main figures are produced directly by [replication-package/JPEReplicationKitFinal/code/analysis/Binscatter_Figures.do](#), [replication-package/JPEReplicationKitFinal/code/analysis/MO_Tightness_Figures.do](#), [replication-package/JPEReplicationKitFinal/code/analysis/Make_CountyMap.py](#), and [replication-package/JPEReplicationKitFinal/code/analysis/Make_BenefitMaps.py](#). A small set of inline policy numbers is derived in [replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do](#).

7.1.2 Data Preparation Code

Program	Dataset or analysis file created	Reproduced?
replication-package/JPEReplicationKitFinal/code/RunBuild.do	Main orchestrator for the full raw-data rebuild and the processed-data build.	Yes
replication-package/JPEReplicationKitFinal/code/build/MakeDataSetsMainPaper_v5_newvac.do	Builds the main analysis panel used in the paper regressions.	Yes
replication-package/JPEReplicationKitFinal/code/build/MakeDataControls.do	Builds the control-variable panel used by the analysis scripts.	Yes
replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Writes the derived policy-scenario CSV that underlies the paper's inline policy numbers.	Yes

7.1.3 Tables and Figures

Output in paper	Program and line number(s)	Reproduced?
Table 1 / tab:Benefits_on_unemp	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Table 2 / tab:Forward_Spec	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py and replication-package/JPEReplicationKitFinal/code/matlab/ProcessQDK.m	Yes
Table 3 / tab:Benefits_on_JobCreation	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Table 4 / tab:Benefits_on_Wages	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Figure 1(a) and Figure 1(b) / fig:binscatter_du and fig:binscatter_qdu	replication-package/JPEReplicationKitFinal/code/analysis/Binscatter_Figures.do	Yes
Figure A-1 / fig:MO_vacPuD	replication-package/JPEReplicationKitFinal/code/analysis/MO_Tightness_Figures.do	Yes
Figure A-2 / fig:Tightness_MO_Borders	replication-package/JPEReplicationKitFinal/code/analysis/MO_Tightness_Figures.do	Yes
Table A-1 / tab:MO_V_U	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py and replication-package/JPEReplicationKitFinal/code/analysis/MO_Tightness_Figures.do	Yes
Figure A-3 / fig:initial-guess	replication-package/JPEReplicationKitFinal/	No

Output in paper	Program and line number(s)	Reproduced?
	code/tex/make_tables.py and replication- package/JPEReplicationKitFinal/ code/matlab/RunMonteCarlo.m	
Table A-2 / tab:Monte-Carlo-Results	replication- package/JPEReplicationKitFinal/ code/tex/make_tables.py and replication- package/JPEReplicationKitFinal/ code/matlab/RunMonteCarlo.m	Yes
Table A-3 / tab:Benefits_on_unemp_OLS	replication- package/JPEReplicationKitFinal/ code/tex/make_tables.py	Yes
Table A-4 / tab:Endog_Vars	replication- package/JPEReplicationKitFinal/ code/tex/make_tables.py and replication- package/JPEReplicationKitFinal/ code/analysis/Endog_Vars.do	Yes
Figure A-4 / fig:endogeneity_test_unemp	replication- package/JPEReplicationKitFinal/ code/analysis/Derived_Policy_C alcs.do	No
Table A-7 / tab:Mobility_LODES	replication- package/JPEReplicationKitFinal/ code/tex/make_tables.py and replication- package/JPEReplicationKitFinal/ code/analysis/Mobility_LODES. o	Yes
Table A-8 / tab:Imputed_Results	replication- package/JPEReplicationKitFinal/ code/tex/make_tables.py and replication- package/JPEReplicationKitFinal/ code/analysis/Imputed_Results. do	Yes

Output in paper	Program and line number(s)	Reproduced?
Table A-9 / app_tab:A-1	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py and replication-package/JPEReplicationKitFinal/code/analysis/TableA2.do	Yes
Table A-11 / tab:Benefits_on_unemp_beg	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Table A-13 / tab:Benefits_on_unemp_SNAP_Mortgage	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Table A-14 / tab:Benefits_on_unemp_stimulus_taxes	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Table A-15 / tab:Benefits_on_unemp_other_policies	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes
Figure A-5 / fig:Counties_Map	replication-package/JPEReplicationKitFinal/code/analysis/Make_CountyMap.py	Yes
Figure A-6 and Figure 6(a-j) / fig:Benefit_Maps and fig:Benefit_Maps_42 through fig:Benefit_Maps_96	replication-package/JPEReplicationKitFinal/code/analysis/Make_BenefitMaps.py	Yes
Table A-16 / tab:app_Benefits_on_Wages	replication-package/JPEReplicationKitFinal/code/tex/make_tables.py	Yes

7.1.4 In-Text Numbers

In-text numbers	Program	Line Number	Reproduced?
16,000	replication-package/JPEReplicationKitFinal/code/analysis/Derived	Not identified	Not yet verified

In-text numbers	Program	Line Number	Reproduced?
	_Policy_Calcs.do		
93%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
5%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
2%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
1,172	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
1,132	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
14	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
0 to 18	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
1,113	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
59	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
282 / 9,024	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified

In-text numbers	Program	Line Number	Reproduced?
26 to 99	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
13 or 20	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
4	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
10.7%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
16	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
13	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
3.02 and 2.15	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
5% to 9.6%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
95% to 90.4%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
10%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
0.4 to 0.5	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified

In-text numbers	Program	Line Number	Reproduced?
0.0528	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
73	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
7.3 to 14.6	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
5/27	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
0.78%	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
30	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified
10,000	replication-package/JPEReplicationKitFinal/code/analysis/Derived_Policy_Calcs.do	Not identified	Not yet verified

7.2 Computing Environment of the Replicator

- Nuvolos v16, 64GB RAM
- Stata/MP 19.5
- Matlab R2024b
- Python 3.12.12
- pdfTeX 3.141592653-2.6-1.40.27

8 Replication

The code and data were accessed via the provided replication package. Several modifications were required to execute the code successfully on the Nuvolos platform.

8.1 Fixes Applied

8.1.1 Python Geopandas / Shapefile Error

The following error occurred during the map generation step because the expected shapefile directory was still compressed:

```
==== python maps ====
ERROR 1: PROJ: proj_create_from_database: Open of /usr/local/lib/conda/share/proj failed
Traceback (most recent call last):
  File "/files/JPE-Mitman-2016256/replication-
package/JPEReplicationKitFinal/code/analysis/Make_BenefitMaps.py", line 47, in <module>
    gdf = gpd.read_file(SHP)
          ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/conda/lib/python3.12/site-packages/geopandas/io/file.py", line
316, in _read_file
    return _read_file_pyogrio(
           ^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/conda/lib/python3.12/site-packages/geopandas/io/file.py", line
576, in _read_file_pyogrio
    return pyogrio.read_dataframe(path_or_bytes, bbox=bbox, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/conda/lib/python3.12/site-packages/pyogrio/geopandas.py", line
382, in read_dataframe
    result = read_func(
             ^^^^^^^^^
  File "/usr/local/lib/conda/lib/python3.12/site-packages/pyogrio/raw.py", line 200, in
read
    return ogr_read(
           ^^^^^^^^^
  File "pyogrio/_io.pyx", line 1412, in pyogrio._io.ogr_read
  File "pyogrio/_io.pyx", line 262, in pyogrio._io.ogr_open
pyogrio.errors.DataSourceError: /files/JPE-Mitman-2016256/replication-
package/JPEReplicationKitFinal/code/analysis/../../data/raw/cb_2025_us_county_20m/cb_2025
_us_county_20m.shp: No such file or directory
FAILED: map figures
```

This was resolved by manually unzipping the archive in `data/raw`:

```
unzip cb_2025_us_county_20m.zip -d cb_2025_us_county_20m
```

8.1.2 Python Directory Path Fix

In `make_tables.py` (line 1144), an incorrect directory reference caused an issue. The code was manually corrected from `HERE` to `IN_DIR`:

```
_mc = IN_DIR / "SenseResultsMonteCarlo.csv"
```

8.1.3 PDFLaTeX Error

The script reported a compilation failure:

```
==== pdflatex ====  
FAILED: paper compile
```

However, upon inspection, the PDF compiled successfully despite the terminal flag.

8.2 Findings

8.2.1 General Issues

- **Missing licensed data:** The README states that to reproduce the published vacancy numbers, the Data Editor replaces the synthetic files in `synthetic-data/` with the licensed HWOL data. These licensed files are not present in the package, meaning true reproduction of the vacancy numbers cannot be achieved as provided.
- **Controlled randomness:** No information is provided regarding controlled randomness (e.g., setting a seed) for any stochastic processes in the code.

8.3 Missing Requirements

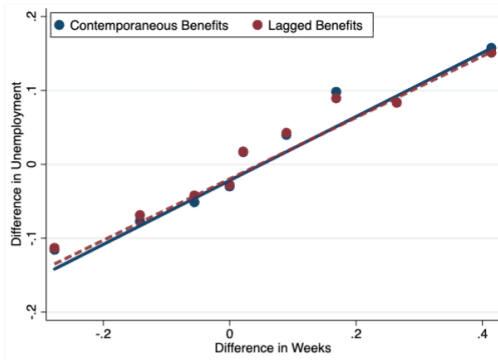
- ☐ Data Availability and Provenance Statements
 - ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
- ☐ Computational requirements
 - ☐ Controlled Randomness (as necessary)
- ☐ List of tables and programs
- ☐ References (Optional, but recommended)

8.4 Data Preparation Code

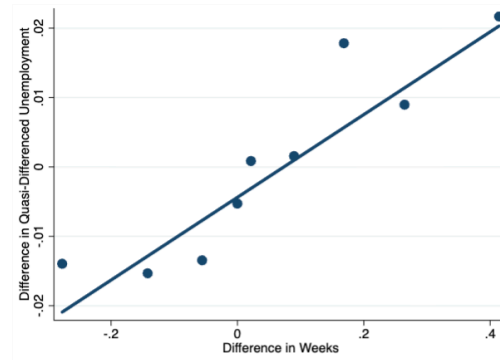
After all modifications, programs ran without errors.

8.5 Tables and Figures

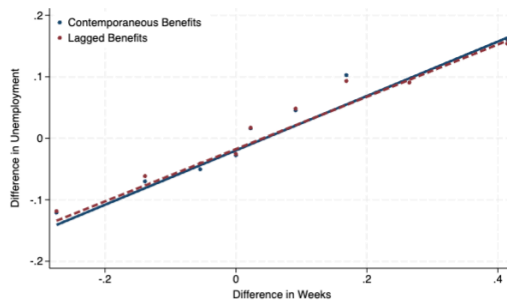
8.5.1 Figures



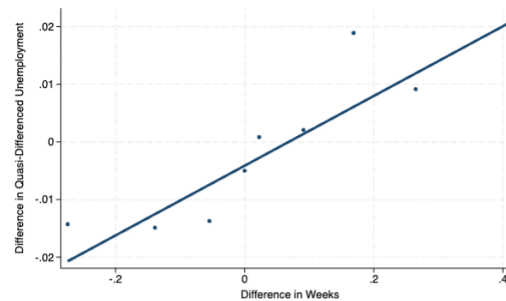
(a) Current and Lagged Benefits and Unemployment



(b) Current Benefits and QD Unemployment

Manuscript

(a) Current and Lagged Benefits and Unemployment

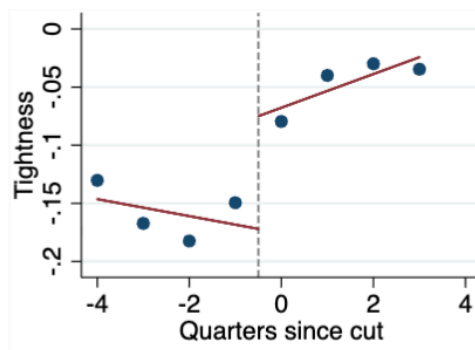


(b) Current Benefits and QD Unemployment

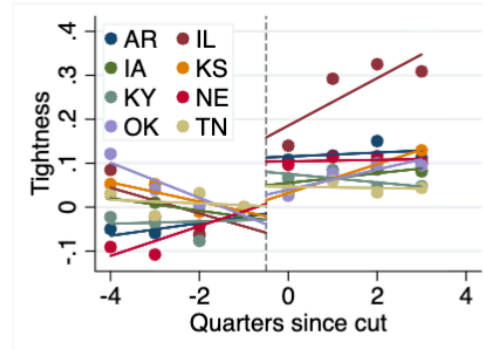
Replication

Figure 1: Figure 1

Finding: Reproduced, but styling is noticeably different from the paper. Variations are present in the grid, the number of scales (horizontal vs. vertical), and the legend formatting.



(a) All Missouri Borders.



(b) Each Missouri Border.

paper

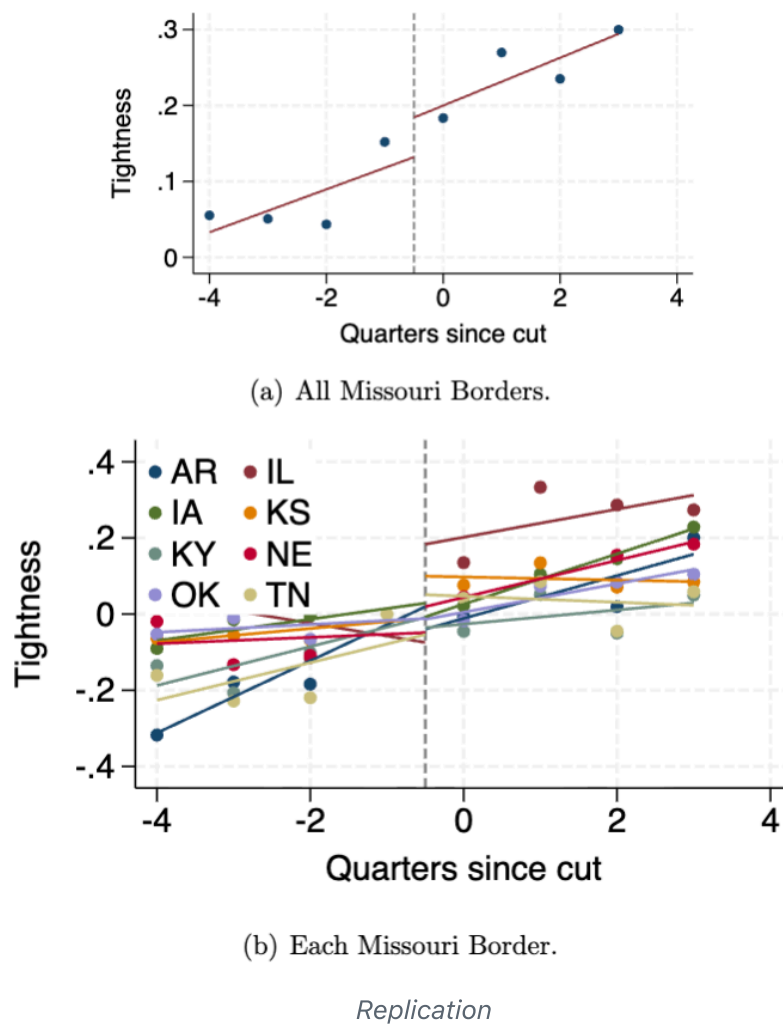


Figure 2: Figure A-1

Finding: Results differ significantly from the paper. Visual styling is also entirely different.

Figure A-3 ([initial_guess.pdf](#))

Finding: Not reproduced by the code.

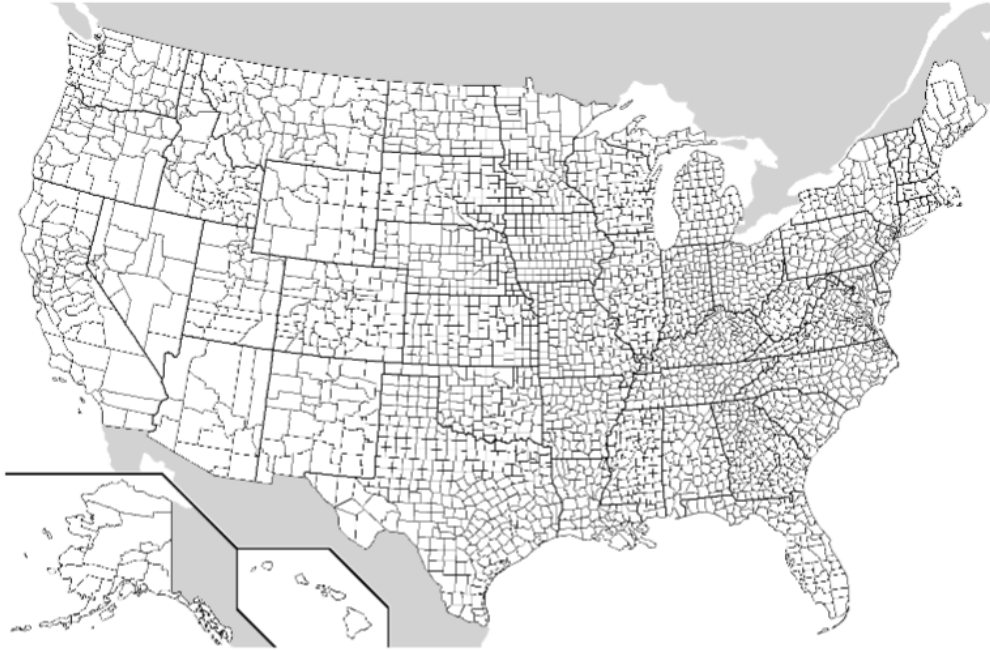


Figure A-5: Map of U.S.A. with state and county outlines.

paper

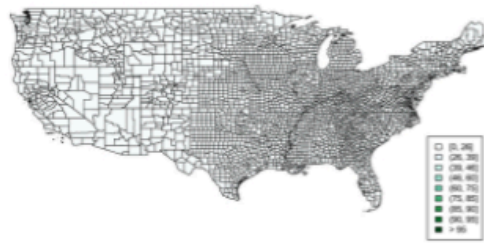


Figure A-5: Map of U.S.A. with state and county outlines.

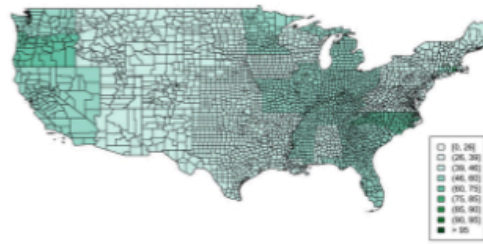
Replication

Figure 3: Figure A-5

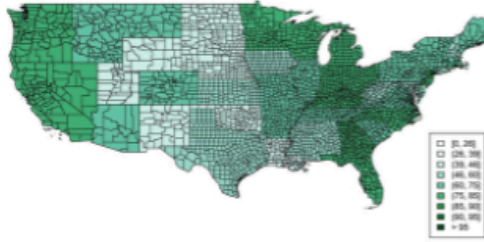
Finding: Reproduced, but features a completely different style and look compared to the published paper.



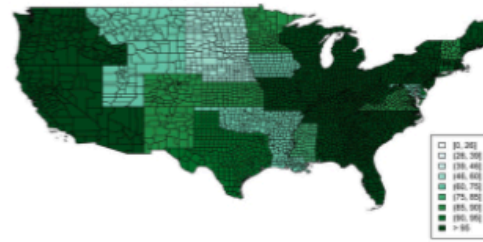
(a) June 2008.



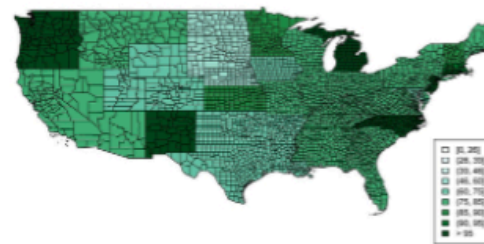
(b) December 2008.



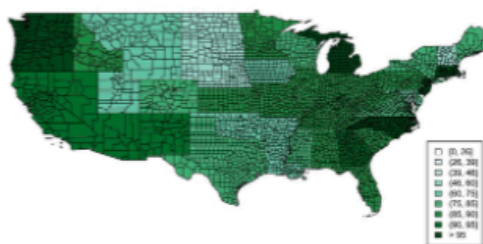
(c) June 2009.



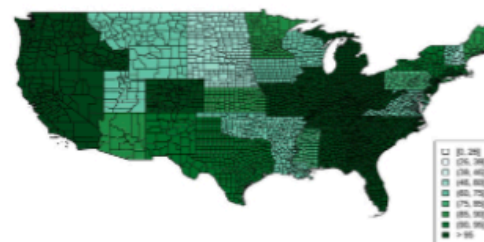
(d) December 2009.



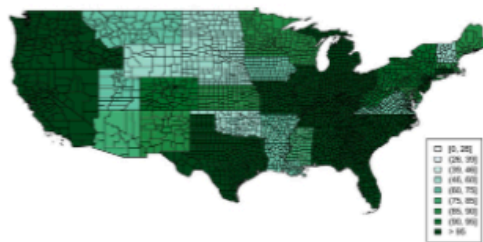
(e) June 2010.



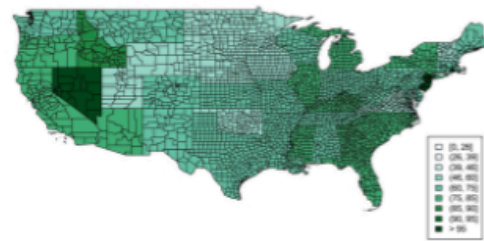
(f) December 2010.



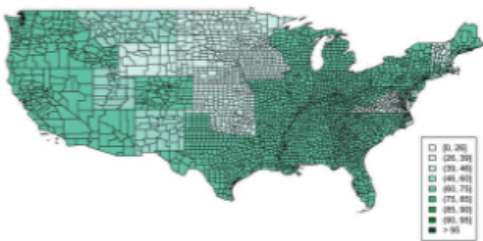
(g) June 2011.



(h) December 2011.



paper



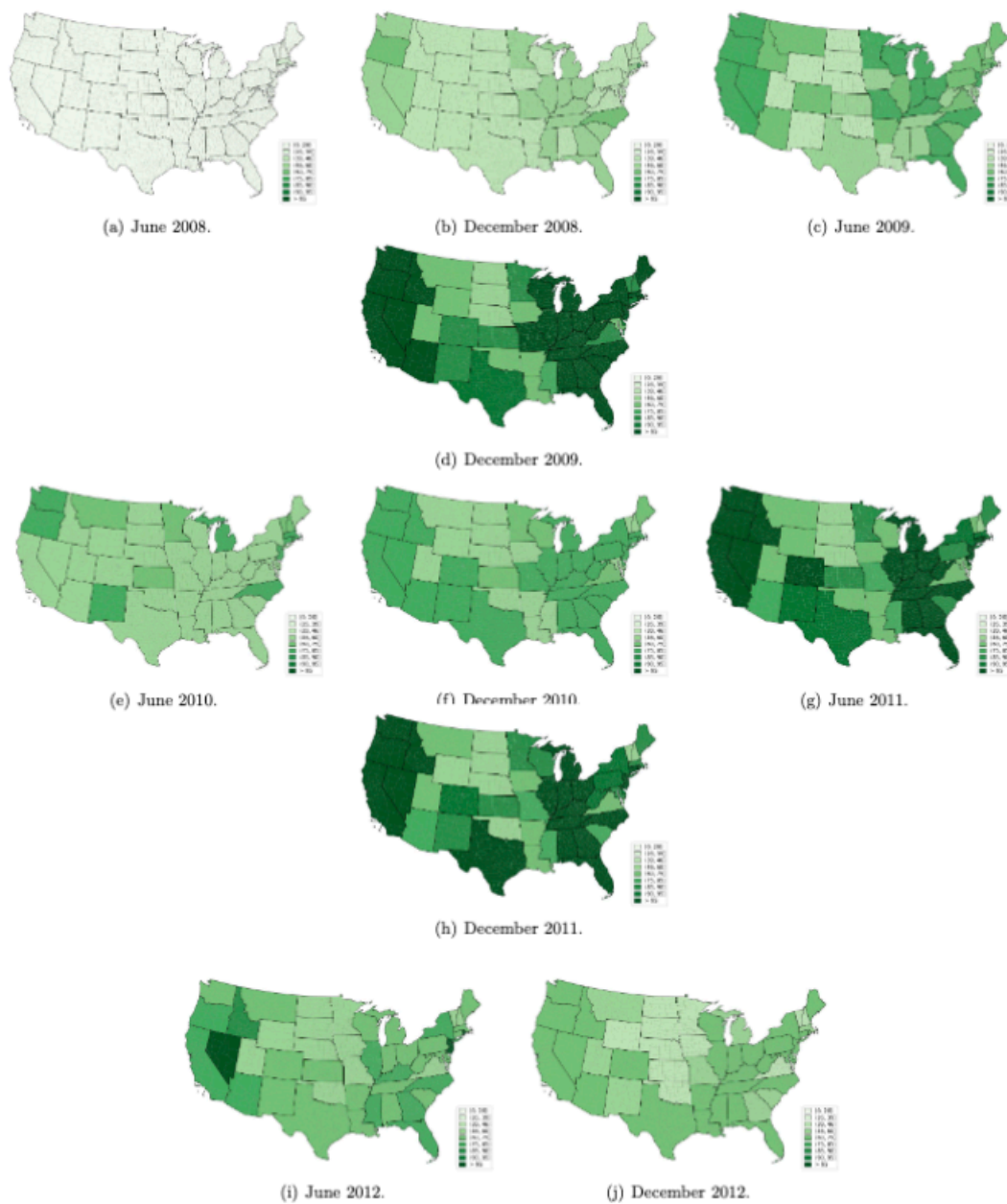


Figure A-6: Unemployment benefit duration across U.S. states during the Great Recession. Selected months.

Replication

Figure 4: Figure A-6

Finding: Reproduced, but features a completely different style and look compared to the published paper.

[StateUCrop.pdf](#)

Finding: Not reproduced by the code.

8.5.2 Tables

Table 1: Table A-1

Table A-1: Changes in Vacancies, Unemployment, and their Ratio along Missouri Borders

Missouri Border with	Change in Tightness	Change in Vacancies	Change in Unemployment
Arkansas	.108	.064	-.045
Illinois	.140	.066	-.073
Iowa	.050	.022	-.028
Kansas	.032	.028	-.004
Kentucky	.067	.043	-.023
Nebraska	.096	.066	-.031
Oklahoma	.026	.016	-.011
Tennessee	.040	.022	-.018

paper

Table A-1: Changes in Vacancies, Unemployment, and their Ratio along Missouri Borders

Missouri Border with	Change in Tightness	Change in Vacancies	Change in Unemployment
Arkansas	-.006	-.050	-.045
Illinois	.135	.062	-.073
Iowa	.022	-.007	-.028
Kansas	.076	.072	-.004
Kentucky	-.045	-.069	-.023
Nebraska	.045	.014	-.031
Oklahoma	-.017	-.028	-.011
Tennessee	.042	.023	-.018

Replication

Finding: The calculated change in vacancies and tightness differ from the reported numbers in the paper.

Table 2: Table A-16

Table A-16: Unemployment Benefit Extensions and Wages

VARIABLES	Job Stayers		New Hires		All Workers	
	Raw	With	Raw	With	Raw	With
	Wages	Tax	Wages	Tax	Wages	Tax
	(1)	(2)	(3)	(4)	(1)	(2)
Weeks of Benefits	0.0232 (0.015)	0.0237 (0.010)	0.0297 (0.030)	0.0299 (0.030)	0.0090 (0.045)	0.0093 (0.045)
N. factors	4	4	1	1	2	2
Observations	25,940	25,940	30,762	30,762	36,460	36,460
R-squared	0.521	0.521	0.119	0.119	0.434	0.434

Note - *p*-values (in parentheses) calculated via bootstrap. Bold indicates *p* < 0.05.

paper

Table A-16: Unemployment Benefit Extensions and Wages

VARIABLES	Job Stayers		New Hires		All Workers	
	Raw	With	Raw	With	Raw	With
	Wages (1)	Tax (2)	Wages (3)	Tax (4)	Wages (1)	Tax (2)
Weeks of Benefits	0.0232 (0.015)	0.0237 (0.000)	0.0297 (0.045)	0.0299 (0.025)	0.0091 (0.035)	0.0093 (0.020)
N. factors	4	4	1	1	2	2
Observations	25,940	25,940	30,762	30,762	36,492	36,492
R-squared	0.521	0.521	0.119	0.119	0.434	0.434

Note - *p*-values (in parentheses) calculated via bootstrap. Bold indicates $p < 0.05$.

Replication

Finding: The replicated table yields a different number of observations and different overall results compared to the paper.

The same issue also appears for Tables A-2, A-3, A-4, A-7, A-11, and A-12, which show discrepancies relative to the published tables.

Table 3: Table A-2

Table A-2: Monte Carlo Results

Parameterization					
Serial, ρ^t	Spatial, ρ^s	True value	Mean	Median	Δ
Panel A: Benchmark Specification					
0.08	0.56	0.0532	0.0531	0.0533	-0.0001
Panel B: Changing Serial Correlation Only					
0.008 + 3 <i>S.E.</i>	0.56	0.0532	0.0535	0.0537	0.0003
0.008 - 3 <i>S.E.</i>	0.56	0.0532	0.0531	0.0533	-0.0001
Panel C: Changing Spatial Correlation Only					
0.08	0.56 + 3 <i>S.E.</i>	0.0532	0.0533	0.0532	0.0001
0.08	0.56 - 3 <i>S.E.</i>	0.0532	0.0526	0.0525	-0.0006
Panel D: Changing Serial and Spatial Correlation					
0.008 + 3 <i>S.E.</i>	0.56 + 3 <i>S.E.</i>	0.0532	0.0535	0.0537	0.0003
0.008 - 3 <i>S.E.</i>	0.56 + 3 <i>S.E.</i>	0.0532	0.0531	0.0532	-0.0001
0.008 + 3 <i>S.E.</i>	0.56 - 3 <i>S.E.</i>	0.0532	0.0533	0.0534	0.0001
0.008 - 3 <i>S.E.</i>	0.56 - 3 <i>S.E.</i>	0.0532	0.0526	0.0526	-0.0006

Note - The table collects the results of all Monte Carlo experiments described in Appendix III.3.2.

paper

Table A-2: Monte Carlo Results

Parameterization		True value	Mean	Median	Δ
Serial, ρ^t	Spatial, ρ^s				
Panel A: Benchmark Specification					
0.08	0.56	0.0532	0.0534	0.0536	0.0002
Panel B: Changing Serial Correlation Only					
0.08 + 3 S.E.	0.56	0.0532	0.0534	0.0534	0.0002
0.08 - 3 S.E.	0.56	0.0532	0.0532	0.0533	-0.0000
Panel C: Changing Spatial Correlation Only					
0.08	0.56 + 3 S.E.	0.0532	0.0530	0.0528	-0.0001
0.08	0.56 - 3 S.E.	0.0532	0.0534	0.0532	0.0002
Panel D: Changing Serial and Spatial Correlation					
0.08 + 3 S.E.	0.56 + 3 S.E.	0.0532	0.0534	0.0537	0.0002
0.08 - 3 S.E.	0.56 + 3 S.E.	0.0532	0.0532	0.0534	0.0001
0.08 + 3 S.E.	0.56 - 3 S.E.	0.0532	0.0530	0.0531	-0.0002
0.08 - 3 S.E.	0.56 - 3 S.E.	0.0532	0.0530	0.0532	-0.0002

Note - The table collects the results of all Monte Carlo experiments described in Appendix III.3.2.

Replication

Table 4: Table A-3

Table A-3: Unemployment Benefit Extensions and Unemployment: Results Based on Additive Effects Specification

VAR.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Weeks of Benefits	0.040 (0.012)	0.037 (0.013)	0.107 (0.004)	0.097 (0.004)	0.042 (0.013)	0.039 (0.012)	0.039 (0.012)	0.037 (0.013)	0.039 (0.014)	0.046 (0.013)
State GDP per Worker		-0.074 (0.038)		-0.140 (0.010)						
Obs.	37,496	37,496	37,401	37,401	35,448	18,840	19,292	24,320	37,496	35,032
R-squared	0.056	0.056	0.095	0.095	0.053	0.065	0.058	0.047	0.056	0.055

Note - Clustered standard errors in parentheses. Bold indicates $p < 0.01$.

paper

Table A-3: Unemployment Benefit Extensions and Unemployment: Results Based on Additive Effects Specification

VAR.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Weeks of Benefits	0.040 (0.012)	0.037 (0.013)	0.106 (0.004)	0.096 (0.004)	0.042 (0.013)	0.044 (0.012)	0.039 (0.012)	0.037 (0.013)	0.039 (0.014)	0.046 (0.013)
State GDP per Worker		-0.080 (0.038)		-0.151 (0.010)						
Obs.	37,496	37,496	37,496	37,496	35,448	18,328	19,292	24,320	37,496	35,032
R-squared	0.056	0.056	0.095	0.096	0.053	0.059	0.058	0.047	0.056	0.055

Note - Clustered standard errors in parentheses. Bold indicates $p < 0.01$.

Replication

Table 5: Table A-4

Table A-4: Unemployment Benefit Extensions and Unemployment, Controlling for County-Level House Prices, Income, and Debt

VARIABLES	Factor Model	OLS	IV-GLS
	(1)	(2)	(3)
Weeks of Benefits	0.059 (0.000)	0.045 (0.000)	0.054 (0.000)
N. factors	2	—	—
Observations	23,307	23,293	4,155
R-squared	0.480	0.074	-0.015

Note - p -values (in parentheses) calculated via bootstrap. Bold indicates $p < 0.01$.

paper

Table A-4: Unemployment Benefit Extensions and Unemployment, Controlling for County-Level House Prices, Income, and Debt

VARIABLES	Factor Model	OLS	IV-GLS
	(1)	(2)	(3)
Weeks of Benefits	0.059 (0.000)	0.044 (0.000)	0.046 (0.001)
N. factors	2	—	—
Observations	23,332	23,318	4,216
R-squared	0.479	0.073	-0.017

Note - p -values (in parentheses). Bold indicates $p < 0.01$.

Replication

Table 6: Table A-7

Table A-6: Estimated Permanent Effect of a 10 Week Benefit Extension from Regressions Coefficients in Model Generated Data

VARIABLES	Unemp. (1)	Vacancies (2)	Tightness (3)
Data	0.159	-0.155	-0.301
Model	0.157	-0.133	-0.279

paper

Table A-6: Estimated Permanent Effect of a 10 Week Benefit Extension from Regressions Coefficients in Model Generated Data

VARIABLES	Unemp. (1)	Vacancies (2)	Tightness (3)
Data	0.159	-0.099	-0.265
Model	0.157	-0.133	-0.279

Replication

Table 7: Table A-11

Table A-11: Unemployment Benefit Extensions and Unemployment: QWI Separations

VAR.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Weeks of Benefits	0.055 (0.000)	0.053 (0.000)	0.120 (0.000)	0.116 (0.000)	0.053 (0.000)	0.061 (0.000)	0.052 (0.000)	0.053 (0.000)	0.054 (0.000)	0.046 (0.000)
State GDP per Worker		0.008 (0.230)		-0.071 (0.0)						
N. factors	2	2	3	3	2	2	2	2	2	2
Obs.	37,496	37,496	37,496	37,496	35,448	18,840	19,292	24,320	37,496	35,032
R-squared	0.449	0.449	0.631	0.632	0.443	0.425	0.416	0.454	0.449	0.452

Note - p -values (in parentheses) calculated via bootstrap. Bold indicates $p < 0.01$.

paper

Table A-11: Unemployment Benefit Extensions and Unemployment: QWI Separations

VAR.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Weeks of Benefits	0.055 (0.000)	0.054 (0.000)	0.096 (0.000)	0.089 (0.000)	0.053 (0.000)	0.067 (0.000)	0.052 (0.000)	0.053 (0.000)	0.054 (0.000)	0.049 (0.000)
State GDP per Worker		0.007 (0.350)		-0.108 (0.000)						
N. factors	2	2	–	–	2	2	2	2	2	2
Obs.	37,496	37,496	–	–	35,448	18,328	19,292	24,320	37,496	35,032
R-squared	0.449	0.449	–	–	0.443	0.408	0.416	0.454	0.449	0.451

Note - p -values (in parentheses) calculated via bootstrap. Bold indicates $p < 0.01$.

Replication

Table 8: Table A-12

Table A-12: Unemployment Benefit Extensions and Job Creation: QWI Separations

VARIABLES	Vacancies (1)	Tightness (2)	Employment (3)
Weeks of Benefits	-0.053 (0.000)	-0.103 (0.000)	-0.006 (0.010)
N. factors	1	1	4
Observations	34,501	34,501	36,779
R-squared	0.104	0.100	0.956

Note - *p*-values (in parentheses) calculated via bootstrap.

Bold indicates $p < 0.01$.

paper

Table A-12: Unemployment Benefit Extensions and Job Creation: QWI Separations

VARIABLES	Vacancies (1)	Tightness (2)	Employment (3)
Weeks of Benefits	-0.034 (0.000)	-0.092 (0.000)	-0.006 (0.000)
N. factors	1	1	4
Observations	37,496	37,496	36,779
R-squared	0.117	0.219	0.956

Note - *p*-values (in parentheses) calculated via bootstrap.

Bold indicates $p < 0.01$.

Replication

8.6 Other issues

- As mentioned before, please correct data citations.
- Please add information on controlled randomness.
- Make sure that when unzipping your replication package, the content is directly in **replication-package** folder. Now, it is double zipped under **JPEReplicationKitFinal**.

9 Classification

Full reproduction can include a small number of apparently insignificant changes in the numbers in the table. Full reproduction also applies when changes to the programs needed to be made, but were successfully implemented.

Partial reproduction means that a significant number (>25%) of programs and/or numbers are different.

Note that if some data is confidential and not available, then a partial reproduction applies. This should be noted in the Reasons.

Note that when all data is confidential, it is unlikely that this exercise should have been attempted.

Failure to reproduce: only a small number of programs ran successfully, or only a small number of numbers were successfully generated (<25%). This also applies when all data is restricted-access and none of the **main** tables/figures are run.

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

9.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☒ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements